Embedded Systems Development

Exercise 3

1. Design a "click counter" using ATmega328P microcontroller and give your design in a schematic diagram.

   The "click counter" should count the number of times that you press a pushbutton switch and display that number in binary using LEDs. It should initially display 0 on the LEDs by turning all LEDs off. You can then start entering a number by clicking the pushbutton switch. When you notify the "click counter" that you are done entering the number by pressing a second pushbutton switch, it should display the number you entered in binary on the LEDs. The number should remain on the LEDs until you press the second pushbutton again to clear it by turning all LEDs off. The "click counter" should be able to count and display any number between 0 and 255.

   As the pins 2 and 3 of ATmega328P is multiplexed for receiving and transmitting data, when the microcontroller serially communicates, they cannot be used as General Purpose Input/Output (GPIO) pins. Therefore, you should reserve them when designing the "click counter" as they would be used for serial communication in a future extension.

2. Write the required firmware for the "click counter" in C.